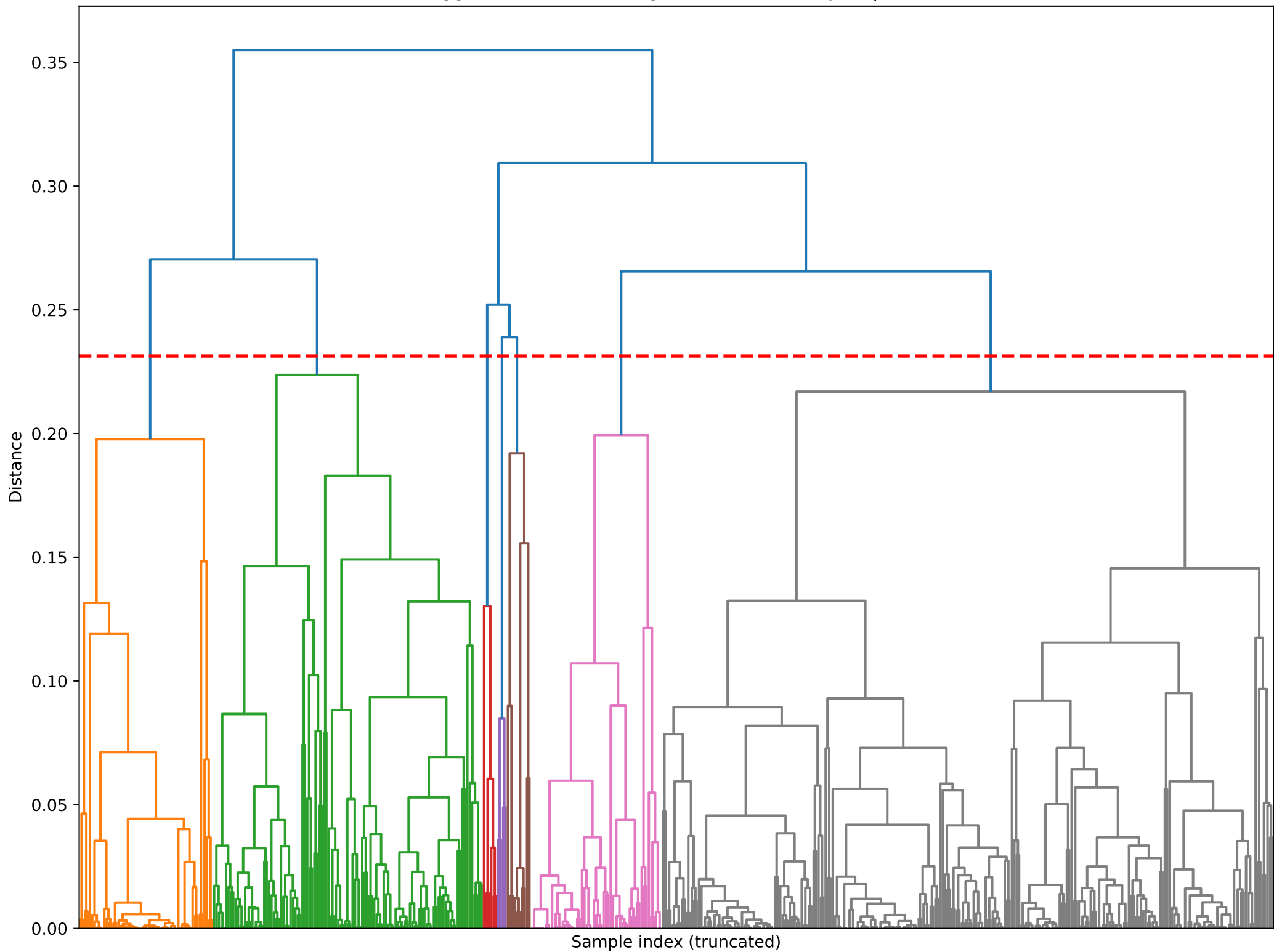


Introduction — experimental setup

Pipeline overview (action → purpose):

- 1) Load `5\_test\_candidate\_cleaned\_final.csv` and validate key column names (e.g., EXAM_SUBJECT, CREDENTIAL_DATE, SCORE_MISSING) → ensure the Python pipeline matches the R reference inputs.
- 2) Aggregate exam-attempt rows to one row per `CANDIDATE\_ID` (scores, pass $\geq$ 145, college $\geq$ 165, avg_retakes formula; subject means only if attempted; missing subjects stay NA) → create a candidate-level clustering dataset.
- 3) Derive profiling-only fields (e.g., age, census_region, race_category and *_label) → support interpretation tables without affecting clusters.
- 4) Completeness Stage 1: treat sentinel 0 as missing for demographics; compute completeness across 26 key fields; keep candidates with completeness ≥ 0.70 → remove records with too little usable information.
- 5) Completeness Stage 2: drop candidates with no demographic identity (no gender AND no race AND no ethnicity) → remove un-profileable cases.
- 6) Variance filter on numeric clustering columns (relative variance ≥ 0.15 ; protect `avg\_score`) → drop near-constant features with no segmentation power.
- 7) Correlation filter on retained numeric columns ($|r| > 0.80$ drop one; protect `avg\_score`) → reduce redundancy / multicollinearity.
- 8) Preprocess for mixed data: continuous vs binary; continuous z-score; non-subject NA → 0, subject NA kept; binary stays 0/1 → align scales while preserving missingness semantics.
- 9) Compute Gower distance on clustering features only (exclude IDs and all demographic/geography) → measure dissimilarity for mixed types.
- 10) Run agglomerative hierarchical clustering (average linkage); choose k via silhouette (2..8) and dendrogram → select an interpretable, data-supported partition.
- 11) Output cluster labels and post-hoc tables (performance, subject, preparation, demographics) → describe cluster profiles after clustering.
- 12) Scope decisions: do NOT implement DIANA; do NOT use study/reason/enrollment as clustering features (profiling only) → avoid mixing survey behavior fields into segmentation.

Agglomerative dendrogram (n=500 sample) | k=7



Using SELECTED_K and final cluster sizes

Using SELECTED_K: 7

Final cluster sizes:
cluster

1	415
2	709
3	428
4	78
5	473
6	1187
7	959

Name: count, dtype: int64

Post-hoc summary — Performance (clustering features)

cluster	N	Avg_Score	Pass_Rate	College_Ready	Credential_Earned	Avg_Retakes	Pct_Online	Used_GED_Ready	Has_Prep_Center
1	415	157.1162	93.8%	19.9%	100.0%	0.1024	0.0%	0.0%	12.3%
2	709	156.1835	88.9%	20.2%	100.0%	1.2997	15.7%	100.0%	0.0%
3	428	154.5370	87.5%	14.4%	100.0%	1.4733	9.7%	100.0%	100.0%
4	78	171.2503	98.8%	82.5%	0.0%	0.2201	5.2%	69.2%	3.9%
5	473	143.7273	36.4%	0.3%	0.0%	0.2292	0.0%	0.0%	20.7%
6	1187	145.2774	50.9%	3.6%	0.0%	0.6919	3.7%	100.0%	0.0%
7	959	144.4390	48.7%	4.3%	0.0%	0.7709	3.5%	100.0%	100.0%

Post-hoc summary — Subject scores

cluster	score_math	score_science	score_reasoning	score_social_studies
1	154.9	158.5	157.6	158.7
2	153.7	158.3	156.6	157.8
3	152.8	157.0	155.2	155.3
4	169.1	171.2	169.1	171.8
5	140.6	145.3	144.7	144.6
6	144.0	148.7	144.9	146.8
7	142.0	147.3	144.8	145.3

Post-hoc summary — Preparation behaviors

cluster	studied_for_ged	study_helpful_adult_education_class	study_helpful_online_course_video_study_materials
1	79.5%	24.1%	22.4%
2	81.1%	23.3%	23.3%
3	84.1%	36.2%	28.5%
4	80.8%	10.3%	21.8%
5	81.8%	37.2%	13.3%
6	83.6%	17.3%	16.8%
7	81.5%	25.3%	12.2%

Post-hoc summary — Demographics

cluster	Avg_Age	Male	Female	Black	White	Hispanic
1	27.6651	67.0%	26.3%	17.1%	57.1%	13.7%
2	27.6319	56.4%	33.3%	12.3%	58.7%	17.8%
3	27.5304	41.1%	47.9%	15.4%	58.9%	20.1%
4	27.7564	53.8%	30.8%	6.4%	50.0%	12.8%
5	32.1522	55.6%	39.1%	34.9%	32.1%	23.5%
6	30.5670	38.8%	33.5%	19.6%	35.3%	18.4%
7	29.9823	26.7%	41.6%	15.6%	35.7%	19.7%

Cluster × Feature contribution matrix (%)

	avg_score	pass_rate	college_ready_rate	avg_retakes	credential_earned	pct_online	used_ged_ready	has_prep_center
1	10.4	11.9	6.4	9.9	19.7	5.8	28.6	7.4
2	11.0	12.0	7.9	9.4	23.7	12.9	9.4	13.6
3	8.5	11.6	3.4	12.8	24.3	5.4	9.6	24.4
4	23.6	11.1	38.6	6.5	9.1	0.2	2.7	8.1
5	9.1	13.0	7.9	9.6	13.5	6.9	34.1	5.8
6	11.4	10.6	9.2	2.4	23.3	3.9	16.0	23.1
7	11.5	10.5	6.9	0.0	19.6	3.7	13.5	34.3

Top features per cluster (by contribution %)

=== Top features per cluster (by contribution %) ===

- Cluster 1: used_ged_ready (28.6%), credential_earned (19.7%), pass_rate (11.9%), avg_score (10.4%), avg_retakes (9.9%)
- Cluster 2: credential_earned (23.7%), has_prep_center (13.6%), pct_online (12.9%), pass_rate (12.0%), avg_score (11.0%)
- Cluster 3: has_prep_center (24.4%), credential_earned (24.3%), avg_retakes (12.8%), pass_rate (11.6%), used_ged_ready (9.6%)
- Cluster 4: college_ready_rate (38.6%), avg_score (23.6%), pass_rate (11.1%), credential_earned (9.1%), has_prep_center (8.1%)
- Cluster 5: used_ged_ready (34.1%), credential_earned (13.5%), pass_rate (13.0%), avg_retakes (9.6%), avg_score (9.1%)
- Cluster 6: credential_earned (23.3%), has_prep_center (23.1%), used_ged_ready (16.0%), avg_score (11.4%), pass_rate (10.6%)
- Cluster 7: has_prep_center (34.3%), credential_earned (19.6%), used_ged_ready (13.5%), avg_score (11.5%), pass_rate (10.5%)